

ECG Anomaly Detection — Session Report

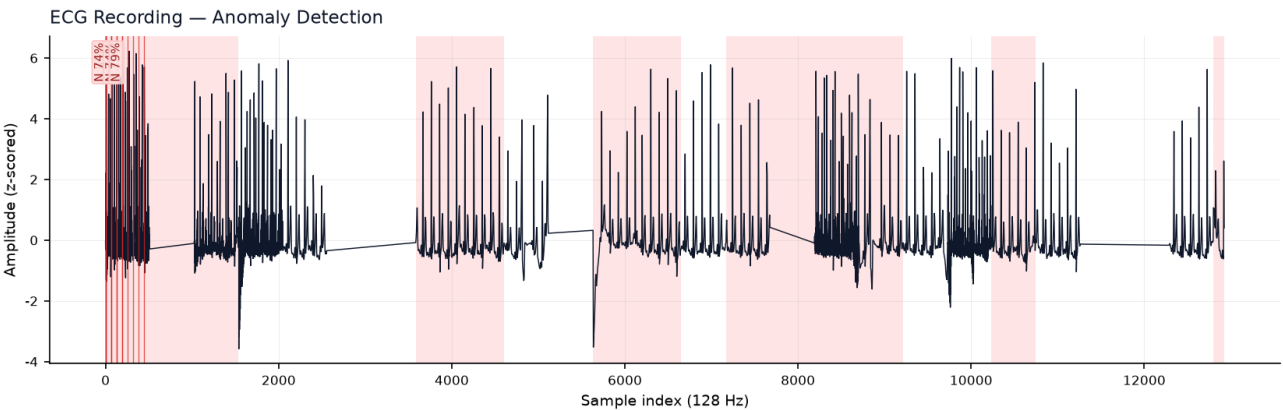
Prepared for: p • Session ID: 00f9dab6 • Generated: 2026-07-05 06:34:11

Session Information

Patient / User	Sriyantha Sadaruwan
Source	mit-bih-arrhythmia (116)
Started	2026-07-05 06:34:11
Ended	2026-07-05 06:35:25
Duration	74.4 s
Status	completed
Total beats analyzed	37,376
Anomalous beats	23,578 (63.08%)
Model	ecg-ae-v2 (v1783189189) • Dilated-U-Net-Autoencoder
Detection threshold (MAE)	0.1341

ECG Recording & Detection Results

The following plot shows the recorded ECG signal (z-score normalized). Red-shaded regions indicate samples flagged as anomalous by the autoencoder (reconstruction error exceeding the trained threshold).



Main Arrhythmia Finding

Premature Ventricular Contraction was the main high-confidence arrhythmia type detected in this session. It appeared 40 time(s) with average confidence 95.7% and maximum confidence 98.3%. Reason: this type is shown because the autoencoder first flagged the ECG segment as anomalous by reconstruction error, and the classifier then assigned this arrhythmia label with at least 50% confidence. Lower-confidence labels are excluded from this report to avoid over-interpreting uncertain model output. **Example:** time 2026-07-05 06:34:37, anomaly score 0.0689, threshold 0.0568, classifier confidence 98.3%.

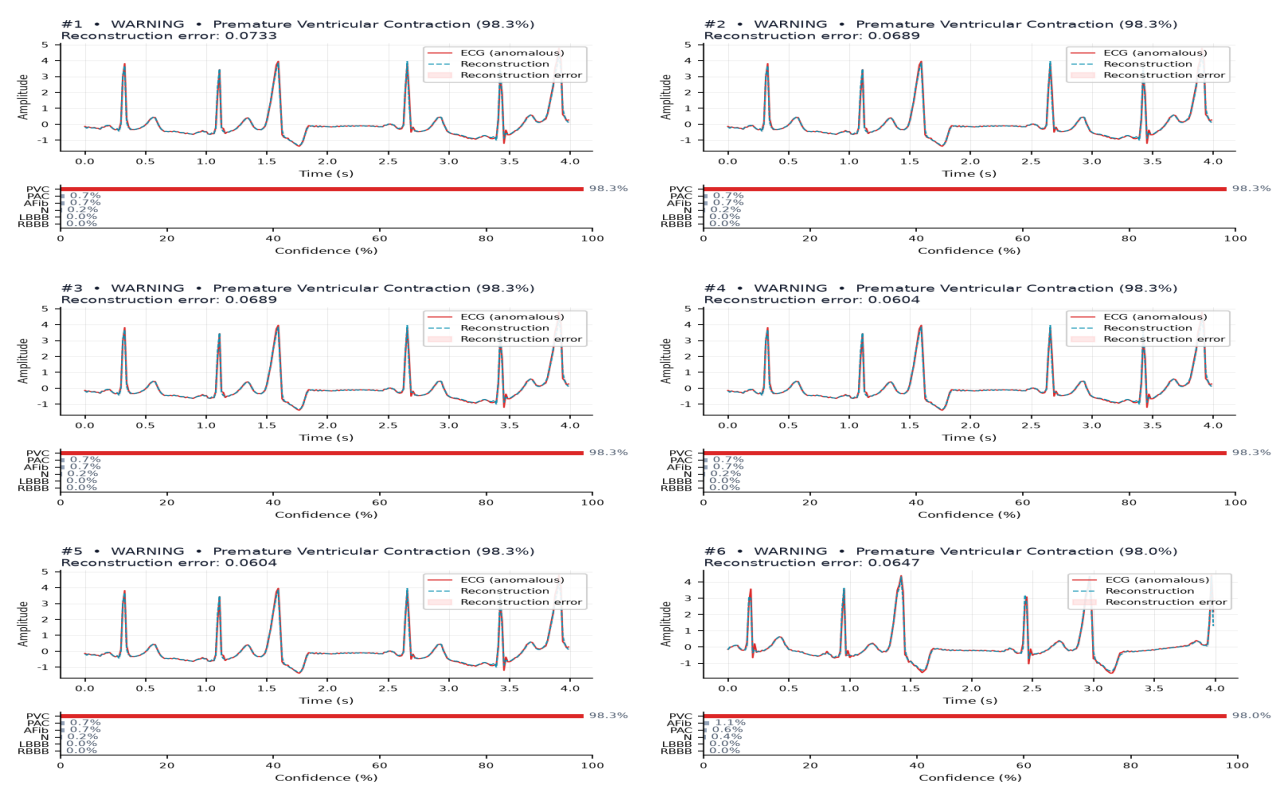
Clinical meaning: A Premature Ventricular Contraction (PVC) is an early heartbeat that originates in the ventricles (the heart's lower chambers) instead of the SA node. The beat comes earlier than expected and is often followed by a brief pause, which can feel like a 'skipped beat' or a 'thud' in the chest.

Why it matters: Occasional PVCs (a few per day) are very common and usually harmless, even in healthy people. They occur in up to 75% of the general population. However, frequent PVCs (more than 10,000 per day) or PVCs occurring in patterns (couplets, bigeminy) in people with structural heart disease may increase the risk of more serious arrhythmias.

Arrhythmia Type	Detections	Avg Confidence	Example
Premature Ventricular Contraction	40	95.7%	score 0.0689, conf 98.3%
Atrial Fibrillation	11	58.6%	score 0.0598, conf 61.2%

Detected Anomaly Waveforms

The following plots show example ECG segments (red) that triggered alerts, alongside the autoencoder's reconstruction of what it expected to see (blue dashed). The shaded region between the two curves represents the reconstruction error that exceeded the detection threshold. When available, examples are selected from arrhythmia classifications above 50% confidence.



Clinical Summary

63.1% of analyzed samples were flagged as anomalous, suggesting sustained abnormal ECG morphology. Urgent cardiology review is recommended. High-confidence classifier findings in this report include: Premature Ventricular Contraction, Atrial Fibrillation. Only arrhythmia labels with confidence above 50% are included.

Physician Notes

Physician signature: _____

Date: _____

This report is generated automatically by an unsupervised deep-learning anomaly detection system and is intended for decision support only. It does not constitute a medical diagnosis and should not replace clinical judgment by a qualified physician.